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| United States Patent | 12403350           |
| Kind Code            | B2                 |
| Date of Patent       | September 02, 2025 |
| Inventor(s)          | Wang; Danni        |

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### Dumbbell and fitness equipment

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#### Abstract

The present disclosure relates to a dumbbell, which includes a first grip rod and two counterweight assemblies, each of both ends of the first grip rod defines a first screw hole, each of the two counterweight assemblies comprises at least one first counterweight, the first counterweight comprises a first connection portion, the first connection portion is threadedly received in the first screw hole to connect the first grip rod with the first counterweight. The present disclosure further provides a fitness equipment applying the dumbbell.

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| <b>Family ID:</b> | <b>85288573</b>                 |
| <b>Appl. No.:</b> | <b>17/863355</b>                |
| <b>Filed:</b>     | <b>July 12, 2022</b>            |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20230067188 A1          | Mar. 02, 2023           |

#### Related U.S. Application Data

continuation-in-part parent-doc US 17403839 20210816 US 11458348 child-doc US 17863355

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#### Publication Classification

**Int. Cl.:** A63B21/072 (20060101)

**U.S. Cl.:**

**Field of Classification Search**

**CPC:**    A63B (21/0722); A63B (21/0726); A63B (21/0728); A63B (21/075); A63B (21/004);  
          A63B (21/072); A63B (21/0724)

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**Background/Summary**

**CROSS-REFERENCE TO RELATED APPLICATIONS** (1) This application is a continuation-in-part of U.S. application Ser. No. 17/403,839, filed on Aug. 16, 2021, the contents of which are incorporated by reference herein.

**FIELD**

(1) This disclosure relates to a technical field of fitness equipment, specifically to a dumbbell and a fitness equipment.

**BACKGROUND**

(2) In order to keep healthy, people use different fitness equipment for exercise, and dumbbells are relatively common fitness equipment. A dumbbell includes a first grip rod and counterweights, and there are multiple counterweights with the same or different weights, so that when the user works out, different users choose the counterweights with different weights to assemble with the first grip rod and then exercise.

(3) However, the replacement operation of dumbbells in the prior art are complex, which involves firstly removing the back fasteners, dismounting the counterweights, then replacing to the counterweights of the desired weight, and getting the fasteners tightened. In this case, the assembly steps are complex and may easily cause difficulties in assembly or lead to injury to the user.

**SUMMARY**

(4) The present disclosure provides a dumbbell for solving the problem of acquiring a simple and flexible counterweight assembly of dumbbells.

(5) To achieve the above purpose, the present disclosure provides a dumbbell, which includes a first grip rod, each of both ends of the first grip rod defines a first screw hole; and two counterweight assemblies, each of the two counterweight assemblies includes a first counterweight, the first

- counterweight includes a first connection portion, the first connection portion is threadedly received in the first screw hole to connect the first grip rod with the first counterweight.
- (6) In at least one embodiment, each of the two counterweight assemblies further includes a second counterweight, the second counterweight includes a second connection portion, a second screw hole is defined in the first counterweight, the second connection portion is threadedly received in the second screw hole.
- (7) In at least one embodiment, no screw hole is defined in an opposite surface of the second counterweight away from the first grip rod.
- (8) In at least one embodiment, the opposite surface of the second counterweight away from the first grip rod is flat.
- (9) In at least one embodiment, a quantity of the at least one first counterweight is more than one, each of the at least one first counterweight defines a second screw hole, two adjacent first counterweights are connected by screwing the first connection portion into respective second screw hole.
- (10) In at least one embodiment, the dumbbell further includes at least two protecting elements, each protecting element is sleeved on respective first counterweight adjacent to the first grip rod; and/or each protecting element is sleeved on respective first counterweight farthest from the first grip rod.
- (11) In at least one embodiment, the protecting element includes a protecting portion, sleeved on a first periphery surface of the first counterweight; and a clamping portion, connected with the protecting portion at an angle, the clamping portion is attached on a first side surface or a second side surface of the first counterweight, the first side surface is opposite to the second side surface.
- (12) In at least one embodiment, the dumbbell further includes at least two protecting elements, each protecting element is sleeved on respective first counterweight adjacent to the first grip rod; and/or each protecting element is sleeved on respective second counterweight.
- (13) In at least one embodiment, the protecting element includes a protecting portion, sleeved on a first periphery surface of the first counterweight or sleeved on a second periphery surface of the second counterweight; and a clamping portion, connected with the protecting portion at an angle, the clamping portion is attached on a first side surface of the first counterweight or attached on a third side surface of the second counterweight.
- (14) In at least one embodiment, the dumbbell further includes at least one first washer, and the first washer is sleeved on the first connection portion of the first counterweight.
- (15) In at least one embodiment, the dumbbell further includes at least one first washer, and the first washer is sleeved on the second connection portion of the second counterweight.
- (16) In at least one embodiment, the dumbbell further includes at least one second washer, and the second washer is sleeved on the first counterweight.
- (17) In at least one embodiment, the dumbbell further includes at least one second washer, and the second washer is sleeved on the second counterweight.
- (18) In at least one embodiment, the first counterweight includes a chamfer, the second counterweight also includes a chamfer, after the first counterweight is connected with the second counterweight, the two chamfers of the first counterweight and the second counterweight cooperatively form a groove, the dumbbell further includes a third washer, the third washer is received in the groove.
- (19) In at least one embodiment, a surface of the first grip rod is provided with at least one anti-slip groove.
- (20) In at least one embodiment, the first counterweight has a weight of 0.25~1.25 Kg; and/or the first counterweight has a diameter of 63~118 mm.
- (21) In at least one embodiment, the dumbbell further includes a sleeve, the sleeve is sleeved on the first grip rod, and the sleeve is made of foam, rubber, or flexible plastic.
- (22) In at least one embodiment, the sleeve in a natural state has a thickness of 3.5~4.5 mm, and the

sleeve in a compressed state has a thickness of 2~3 mm.

(23) The present disclosure also provides a fitness equipment, which includes a second grip rod and at least two dumbbells, a length of the second grip rod is greater than a length of the first grip rod, and the at least two dumbbells are threadedly connected with two opposite ends of the second grip rod.

(24) In at least one embodiment, the second grip rod is integrally formed; or, the second grip rod is assembled in a detachable manner.

(25) The present disclosure involves beneficial effects that: the first counterweights are connected to the first grip rod through screwing the first connection portions into the first screw holes, the thread structure has a firm connection, thus avoiding the first counterweights getting loosen or shaking, and guaranteeing a high usage safety. Further, the first counterweight can be separated from the first grip rod by unscrewing the first connection portion from the first screw hole, the assembly operation and the disassembly operation are simple and flexible. Therefore, the dumbbell of the present disclosure has the advantages of safety, simple, and flexible.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Implementations of the present disclosure will now be described, by way of embodiment, with reference to the attached figures. It should be understood, the drawings are shown for illustrative purpose only, for ordinary person skilled in the art, other drawings obtained from these drawings without paying creative labor by an ordinary person skilled in the art should be within scope of the present disclosure.

(2) FIG. 1 is a schematic structure diagram of a dumbbell according to a first embodiment of the present disclosure.

(3) FIG. 2 is an exploded diagram of the dumbbell of FIG. 1.

(4) FIG. 3 is a cross sectional view of the dumbbell of FIG. 1 along a line

(5) FIG. 4 is a schematic structure diagram of a dumbbell according to a second embodiment of the present disclosure.

(6) FIG. 5 is a schematic structure diagram of a dumbbell according to a third embodiment of the present disclosure.

(7) FIG. 6 is a schematic structure diagram of a dumbbell according to a fourth embodiment of the present disclosure.

(8) FIG. 7 is a schematic structure diagram of a dumbbell according to a fifth embodiment of the present disclosure.

(9) FIG. 8 is a schematic structure diagram of a dumbbell according to a sixth embodiment of the present disclosure.

(10) FIG. 9 is a schematic structure diagram of a dumbbell according to a seventh embodiment of the present disclosure.

(11) FIG. 10 is a schematic structure diagram of a dumbbell according to an eight embodiment of the present disclosure.

(12) FIG. 11 is a schematic structure diagram of a fitness equipment according to a first embodiment of the present disclosure.

(13) FIG. 12 is a schematic structure diagram of a fitness equipment according to a second embodiment of the present disclosure.

(14) FIG. 13 is an exploded diagram of the fitness equipment of FIG. 12.

(15) FIG. 14 is a schematic structure diagram of a fitness equipment according to a third embodiment of the present disclosure.

### DETAILED DESCRIPTION

(16) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the exemplary embodiments described herein. However, it will be understood by those of ordinary skill in the art that the exemplary embodiments described herein may be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the exemplary embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

(17) The term “comprising” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series, and the like. The present disclosure is illustrated by way of example and not by way of limitation in the figures of the accompanying drawings in which like references indicate similar elements. It should be noted that references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and such references can mean “at least one”. In addition, the terms “first” and “second” are used for descriptive purposes only and cannot be understood as indicating or implying relative importance or implying the number of indicated technical features. Thus, the features defined as “first” and “second” may explicitly or implicitly include one or more of the said features. In the description of embodiments of the application, “a plurality of” means two or more, unless otherwise specifically defined.

(18) Referring to FIGS. 1 to 3, in the first embodiment of the present disclosure, the dumbbell **100a** includes a first grip rod **10a** and two counterweight assemblies **30a** symmetrically connected to both ends of the first grip rod **10a**. Each of both ends of the first grip rod **10a** defines a first screw hole **11a**, each counterweight assembly **30a** includes at least one first counterweight **31a**, the first counterweight **31a** includes a first connection portion **311a**, the first connection portion **311a** is threadedly received in the first screw holes **11a** to connect the first grip rod **10a** with the first counterweight **31a**. The first counterweights **31a** are connected to the first grip rod **10a** through screwing the first connection portions **311a** into the first screw holes **11a**, the thread structure has a firm connection, thus avoiding the first counterweights **31a** getting loosen or shaking, and guaranteeing a high usage safety. Further, the first counterweight **31a** can be separated from the first grip rod **10a** by unscrewing the first counterweight **31a** from the first screw hole **1a**, the assembly operation and the disassembly operation are simple and flexible. Therefore, the dumbbell **100a** of the present disclosure has the advantages of safety, simple, and flexible.

(19) In at least one embodiment, the first connection portion **311a** includes an outer thread.

(20) In at least one embodiment, the first connection portion **311a** is arranged on a side of the first counterweight **31a** adjacent to the first grip rod **10a**.

(21) In at least one embodiment, the first connection portion **311a** is integrated with the first counterweight **31a**.

(22) In another embodiment, the first connection portion **311a** and the first counterweight **31a** are formed separately, and then connected together.

(23) In at least one embodiment, the first grip rod **10a** is cylindrical.

(24) In at least one embodiment, a surface of the first grip rod **10a** is provided with at least one anti-slip groove **12a**, so that the flipping through hands will not easily happen when user holds the first grip rod **10a**.

(25) In at least one embodiment, the first screw holes **11a** are communicated with each other, or the first screw holes **11a** are blind holes and spaced apart from each other.

(26) In at least one embodiment, a quantity of the at least one first counterweight **31a** is more than one, each of the at least one first counterweight **31a** defines a second screw hole **313a**, two adjacent first counterweights **31a** are connected by screwing the first connection portion **311a** into

respective second screw hole **313a**. The number of first counterweights **31a** can be increased or decreased infinitely to form the dumbbells **100a** in various weights combination. The first counterweights **31a** can be combined and shared with each other to adjust the weight. In detail, user can assemble at least one first counterweights **31a** with the first grip rod **10a**, or disassemble at last one first counterweights **31a** from the first grip rod **10a**, as the assembly operation and the disassembly operation are simple and flexible.

(27) In at least one embodiment, the second screw hole **313a** is formed at a side of the first counterweight **31a** away from the first grip rod **10a**.

(28) The weight of the dumbbell is generally 2.5 Kg, 5 Kg, 7.5 Kg, 10 Kg, 15 Kg, or 25 Kg. In existing art, each counterweight assembly of the dumbbell includes a connecting counterweight connected with the grip rod, at least one middle counterweight connected with the connecting counterweight, and a lateral counterweight connected with the middle counterweight, the connecting counterweight has a diameter of 63~148 mm and a weight of 0.5~5.75 Kg, the middle counterweight has a diameter of 63~148 mm and a weight of 0.25~4 Kg, the lateral counterweight has a diameter of 63~148 mm, and a weight of 0.25~2.5 Kg, users can assemble the dumbbell with desired weight. The weight of the grip rod may be 0.5 Kg, and the weight of the grip rod can be adjusted according to actual needs. Please referring to table 1, which shows the diameters and weights of the connecting counterweight, the middle counterweight, and the lateral counterweight of the existing dumbbell.

(29) TABLE-US-00001

| TABLE 1 |      | Diameters and weights of the connecting counterweight, the middle counterweight, and the lateral counterweight of the existing dumbbell |      |                      |      |                       |      |                   |      |                   |      |                   |     |
|---------|------|-----------------------------------------------------------------------------------------------------------------------------------------|------|----------------------|------|-----------------------|------|-------------------|------|-------------------|------|-------------------|-----|
|         |      | connecting counterweight                                                                                                                |      | middle counterweight |      | lateral counterweight |      | diameter/ weight/ |      | diameter/ weight/ |      | diameter/ weight/ |     |
|         |      | mm                                                                                                                                      | Kg   | mm                   | Kg   | mm                    | Kg   | mm                | Kg   | mm                | Kg   | mm                | Kg  |
| 63      | 0.5  | 63                                                                                                                                      | 0.25 | 63                   | 0.25 | 83                    | 1    | 83                | 0.75 | 83                | 0.5  | 93                | 1.5 |
| 93      | 1.25 | 93                                                                                                                                      | 0.75 | 98                   | 2.25 | 98                    | 1.5  | 98                | 1    | 118               | 3.25 | 118               | 2.5 |
| 118     | 1.5  | 118                                                                                                                                     | 2.5  | 118                  | 1.5  | 148                   | 5.75 | 148               | 4    | 148               | 2.5  |                   |     |

(30) It should be understood that, the weight of the existing dumbbell can be adjusted according to actual needs. For the existing dumbbell with a maximum weight of 2.5 Kg, its weight can be adjusted to 1 Kg, 1.5 Kg, or 2.5 Kg. For the existing dumbbell with a maximum weight of 5 Kg, its weight can be adjusted to 2.5 Kg, 3.5 Kg, 4 Kg, or 5 Kg. For the existing dumbbell with a maximum weight of 7.5 Kg, its weight can be adjusted to 3.5 Kg, 5 Kg, 6 Kg, or 7.5 Kg. For the existing dumbbell with a maximum weight of 10 Kg, its weight can be adjusted to 5 Kg, 7 Kg, 8 Kg, or 10 Kg. For the existing dumbbell with a maximum weight of 15 Kg, its weight can be adjusted to 7 Kg, 10 Kg, 12 Kg, or 15 Kg. For the existing dumbbell with a maximum weight of 25 Kg, its weight can be adjusted to 12 Kg, 17 Kg, 20 Kg, or 25 Kg. Obviously, the existing dumbbell can be adjusted to have four different weights at most.

(31) In the present disclosure, the first counterweight has a diameter of 63~118 mm and a weight of 0.25~1.25 Kg, the second counterweight has a diameter of 63~118 mm and a weight of 0.25~1 Kg, users can assemble dumbbell with desired weight. The first grip rod **10a** has a weight of 0.5 Kg, and the weight of the first grip rod **10a** can be adjusted according to actual needs. Please referring to table 2, which shows the diameters and weights of the first connecting counterweight and the second counterweight of the dumbbell of the present disclosure.

(32) TABLE-US-00002

| TABLE 2 |      | Diameters and weights of the first counterweight and the second counterweight of the dumbbell of the present disclosure |      |                      |      |                   |      |                   |      |
|---------|------|-------------------------------------------------------------------------------------------------------------------------|------|----------------------|------|-------------------|------|-------------------|------|
|         |      | first counterweight                                                                                                     |      | second counterweight |      | diameter/ weight/ |      | diameter/ weight/ |      |
|         |      | mm                                                                                                                      | Kg   | mm                   | Kg   | mm                | Kg   | mm                | Kg   |
| 63      | 0.25 | 63                                                                                                                      | 0.25 | 98                   | 0.75 | 98                | 0.75 | 118               | 1.25 |
| 118     | 1.25 | 118                                                                                                                     | 1    | 118                  | 1.25 | 118               | 1    | 118               | 1.25 |
| 118     | 1    | 118                                                                                                                     | 1.25 | 118                  | 1    | 118               | 1.25 | 118               | 1    |

(33) The weight and diameter of the first counterweight **31a** are moderate, user can assemble several first counterweights **31a** on the first grip rod **10a** simultaneously, can also disassemble several first counterweights **31a** from the first grip rod **10a** simultaneously. User can assemble several first counterweights **31a** on another one first counterweight **31a**, can also disassemble several first counterweights **31a** from another one first counterweight **31a**. The assembly operation and the disassembly operation are simple and flexible.

(34) In detail, in existing art, manufacturer needs to produce eighteen types of counterweights. In the present disclosure, manufacturer only needs to produce six types of counterweights need to be produced. Obviously, the dumbbell **100a** of the present disclosure has the advantages of simple, economic, and time saving.

(35) It should be understood that, the weight of the dumbbell **100a** of the present disclosure can be adjusted according to actual needs. For the dumbbell **100a** of the present disclosure with a maximum weight of 2.5 Kg, its weight can be adjusted to 1 Kg, 1.5 Kg, 2 Kg, or 2.5 Kg. For the dumbbell **100a** of the present disclosure with a maximum weight of 5 Kg, its weight can be adjusted to 2 Kg, 3.5 Kg, or 5 Kg. For the dumbbell **100a** of the present disclosure with a maximum weight of 7.5 Kg, its weight can be adjusted to 2.5 Kg, 3 Kg, 5 Kg, 5.5 Kg, or 7.5 Kg. For the dumbbell **100a** of the present disclosure with a maximum weight of 10 Kg, its weight can be adjusted to 2.5 Kg, 3 Kg, 5 Kg, 5.5 Kg, 7.5 Kg, or 10 Kg. For the dumbbell **100a** of the present disclosure with a maximum weight of 15 Kg, its weight can be adjusted to 2.5 Kg, 3 Kg, 5 Kg, 5.5 Kg, 7.5 Kg, 10 Kg, 10.5 Kg, 12.5 Kg, 13 Kg, or 15 Kg. Obviously, the dumbbell **100a** of the present disclosure can be adjusted to have ten different weights. The dumbbell **100a** of the present disclosure can be adjusted to have more different weights.

(36) In at least one embodiment, each counterweight assembly **30a** further includes a second counterweight **33a**, each second counterweight **33a** includes a second connection portion **331a**, each second counterweight **33a** is connected with respective first counterweight **31a** by screwing the second connection portion **331a** into respective second screw hole **313a**. The second counterweight **33a** can be screwed with the first counterweight **31a** and unscrewed from the first counterweight **31a**, the assembly operation and the disassembly operation of the second counterweight **33a** are simple and flexible.

(37) In at least one embodiment, the second connection portion **331a** includes an outer thread.

(38) In at least one embodiment, no screw hole is defined in an opposite surface of the second counterweight **33a** away from the first grip rod **10a**.

(39) In at least one embodiment, the opposite surface of the second counterweight **33a** away from the first grip rod **10a** is flat.

(40) In this embodiment, the first counterweight **31a** and the second counterweight **33a** are circular in shape. In other embodiments, the first counterweight **31a** and the second counterweight **33a** may be shaped as polygons such as hexagons to facilitate assembly operations.

(41) In at least one embodiment, a diameter of the second counterweight **33a** is substantially equal to that of the first counterweight **31a**, and a thickness of the second counterweight **33a** is substantially equal to that of the first counterweight **31a**. So that, the assembly operation and the disassembly operation of the first counterweight **31a** may not be affected by the second counterweight **33a**.

(42) In order to facilitate assembly, the first connection portions **311a**, the second connection portions **331a**, the first screw holes **11a**, and the second screw holes **313a** have the same thread structure size.

(43) In at least one embodiment, the dumbbell **100a** further includes at least two protecting elements **50a**, the protecting elements **50a** are sleeved on the first counterweights **31a** adjacent to the first grip rod **10a**. Preferably, each the protecting element **50a** is sleeved on respective first counterweight **31a** closest to the first grip rod **10a**.

(44) It can be seen that the protecting elements **50a** will contact the ground before the first counterweights **31a** and the second counterweight **33a**, the protecting elements **50a** can achieve the effect of isolating the dumbbell **100a** from the ground, reducing the wear of the first counterweights **31a** and the second counterweight **33a**, and as well as avoiding the wear to the ground. The structure of the dumbbell **100a** is simple and practical.

(45) In order to increase the friction of the protecting element **60**, an outer side of the protecting element **60** is provided with a non-slip structure, such as a pattern, knurling, bump structure, etc.

- (46) In at least one embodiment, each protecting elements **50a** is sleeved on respective first counterweight **31a** farthest from the first grip rod **10a**.
- (47) In at least one embodiment, the protecting element **50a** includes a protecting portion **51a**, sleeved on a first periphery surface **315a** of the first counterweight **31a**, and a clamping portion **53a** connected with the protecting portion **51a** at an angle, the clamping portion **53a** is attached on a first side surface **317a** of the first counterweight **3a**.
- (48) In at least one embodiment, the angle between the protecting portion **51a** and the clamping portion **53a** has a range of 70~110°, preferably 90°.
- (49) In at least one embodiment, a width of the protecting portion **51a** is substantially equal to that of the first counterweight **31a**, so as to completely isolating the dumbbell **100a** from the ground.
- (50) In at least one embodiment, the protecting element **50a** is made of rubber, foam, flexible plastic, silicone, or the like.
- (51) In at least one embodiment, the dumbbell **100a** further includes at least one first washer **70a**, and the at least one first washer **70a** is sleeved on the first connection portion **311a** and/or the second connection portion **331a**. The at least one first washer **70a** is clamped between the first counterweight **31a** and the second counterweight **33a**, and/or between the first counterweights **31a**, so as to prevent the user from screwing the threads that hard to cause jamming problem.
- (52) In at least one embodiment, the first washer **70a** is made of rubber, foam, flexible plastic, silicone, or the like.
- (53) Referring to FIG. 4, the present disclosure provides a dumbbell **100b** according to a second embodiment. The dumbbell **100b** is similar to the dumbbell **100a** in structure. The difference between the dumbbell **100b** and dumbbell **100a** includes: the protecting elements **50b** are sleeved on the second counterweights **33b**.
- (54) In at least one embodiment, the protecting portion **51b** is sleeved on the second periphery surface **333b** of the second counterweight **33b**, and the clamping portion **53b** is attached on the third side surface **335b** of the second counterweight **33b**. No matter the dumbbell **100b** is placed horizontally or vertically, the protecting elements **50b** can achieve the effect of isolating the dumbbell **100b** from the ground.
- (55) In at least one embodiment, the third side surface **335b** of the second counterweight **33b** is farthest from the first grip rod **10b**.
- (56) Referring to FIG. 5, the present disclosure provides a dumbbell **100c** according to a third embodiment. The dumbbell **100c** is similar to the dumbbell **100a** in structure. The difference between the dumbbell **100c** and dumbbell **100a** includes: the second counterweights **33c** and the first counterweights **31c** closest to the first grip rod **10c** are sleeved with the protecting elements **50c**.
- (57) In at least one embodiment, the protecting portion **51c** is sleeved on the first periphery surface **315c** of the first counterweight **31c**, and the clamping portion **53c** is attached on the first side surface **317c** of the first counterweight **31c**.
- (58) In at least one embodiment, the protecting portion **51c** is sleeved on the second periphery surface **333c** of the second counterweight **33c**, and the clamping portion **53c** is attached on the third side surface **335c** of the second counterweight **33c**.
- (59) Referring to FIG. 6, the present disclosure provides a dumbbell **100d** according to a fourth embodiment. The dumbbell **100d** is similar to the dumbbell **100c** in structure. The difference between the dumbbell **100d** and dumbbell **100c** includes: the dumbbell **100d** does not include the second counterweights, and the first counterweights **31d** are threadedly connected with the first grip rod **10d**.
- (60) In at least one embodiment, two first counterweights **31d** are respectively connected with two ends of the first grip rod **10d**, respectively. Two protecting elements **50d** are sleeved on the two first counterweights **31d**, respectively.
- (61) In at least one embodiment, each end of the first grip rod **10d** are connected with at least two



first counterweights **31d**, and the first counterweights **31d** closest to the first grip rod **10d** and the first counterweights **31d** farthest to the first grip rod **10d** are sleeved with the protecting elements **50d**.

(62) In at least one embodiment, the protecting portion **51d** is sleeved on the first periphery surface **315d** of the first counterweight **31d**, and the clamping portion **53d** is attached on the first side surface **317d** of the first counterweight **31d**.

(63) In at least one embodiment, the protecting portion **51d** is sleeved on the first periphery surface **315d** of the first counterweight **31d**, and the clamping portion **53d** is attached on the second side surface **319d** of the first counterweights **31d**. The first side surface **317d** is opposite to the second side surface **319d**.

(64) Referring to FIG. 7, the present disclosure provides a dumbbell **100e** according to a fifth embodiment. The dumbbell **100e** is similar to the dumbbell **100b** in structure. The difference between the dumbbell **100e** and dumbbell **100b** includes: the dumbbell **100e** does not include the first counterweights, the second counterweights **33e** are threadedly connected with the first grip rod **10e**.

(65) In at least one embodiment, the second connection portions **331e** are threadedly received in the first screw holes **11e**.

(66) In at least one embodiment, the weight of the dumbbell **100e** of the present disclosure can be adjusted to a minimum of 1 kg.

(67) In at least one embodiment, the first washer **70e** can be arranged between the first grip rod **10e** and the second counterweight **33e**. In detail, the first washer **70e** is sleeved around the second connection portion **331e**, and clamped between the second counterweight **33e** and the first grip rod **10e**.

(68) Referring to FIG. 8, the present disclosure provides a dumbbell **100f** according to a sixth embodiment. The dumbbell **100f** is similar to the dumbbell **100a** in structure. The difference between the dumbbell **100f** and dumbbell **100a** includes: the dumbbell **100f** further includes a sleeve **80f** sleeved on the first grip rod **10f**.

(69) In at least one embodiment, the sleeve **80f** is made of foam, rubber, or flexible plastic.

(70) In at least one embodiment, the sleeve **80f** in a natural state has a thickness of 3.5~4.5 mm, and the sleeve **80f** in a compressed state has a thickness of 2~3 mm, in this case, the sleeve **80f** can provide user a good sense of grip, and users can easily keep their body weight. In detail, the thickness of a compressed portion of the sleeve **80f** is 2~3 mm.

(71) Referring to FIG. 9, the present disclosure provides a dumbbell **100g** according to a seventh embodiment. The dumbbell **100g** is similar to the dumbbell **100a** in structure. The difference between the dumbbell **100g** and dumbbell **100a** includes: the dumbbell **100g** does not include the protecting element, and the dumbbell **100g** further includes a second washer **40g**, and the second washer **40g** is sleeved on the first counterweight **31g** and/or sleeved on the second counterweight **33g**.

(72) In at least one embodiment, the second washer **40g** is ring-shaped.

(73) In at least one embodiment, an outer surface of the second washer **40g** is protruded from the first counterweight **31g** and the second counterweight **33g**. It can be seen that the third washer **40g** will contact the ground before the first counterweights **31g** and the second counterweight **33g**, and can achieve the effect of isolating the dumbbell **100g** from the ground.

(74) Referring to FIG. 10, the present disclosure provides a dumbbell **100h** according to an eighth embodiment. The dumbbell **100h** is similar to the dumbbell **100c** in structure. The differences between the dumbbell **100h** and dumbbell **100c** include: the dumbbell **100h** does not include the protecting element, the first counterweight **31h** includes a chamfer **312h**, the second counterweight **31h** also includes a chamfer **312h**, after the first counterweight **31h** is connected with the second counterweight **31h**, the two chamfers **312** of the first counterweight **31h** and the second counterweight **31h** cooperatively form a groove **314h**, the dumbbell **100h** further includes a third

washer **60h**, the third washer **60h** is received in the groove **314h**.

(75) In at least one embodiment, a second side surface **319h** of the first counterweight **31h** includes defines the chamfer **312h**, a fourth side surface **337h** of the second counterweight **33h** also defines the a chamfer **312h**, the fourth side surface **337h** of the second counterweight **33h** is adjacent to the second side surface **319h** of the first counterweight **31h**, after the first counterweight **31h** is connected with the second counterweight **33h**, the chamfers **312h** cooperatively form the groove **314h**.

(76) In at least one embodiment, the third washer **60h** is ring-shaped.

(77) In at least one embodiment, an outer surface of the third washer **60h** is protruded from the first counterweight **31h** and the second counterweight **33h**.

(78) In at least one embodiment, the second side surface **319h** of the first counterweight **31h** includes a chamfer **312h**, and the first side surface **315h** of the first counterweight **31h** includes a chamfer **312h**, after the two adjacent first counterweights **31h** are connected together, the two chamfers **312h** cooperatively form another groove **314h**, the third washer **60h** may be received in the groove **314h**.

(79) It can be seen that the third washer **60h** will contact the ground before the first counterweights **31h** and the second counterweight **33h**, and can achieve the effect of isolating the dumbbell **100h** from the ground.

(80) Referring to FIG. **11**, the present disclosure provides a fitness equipment **200** according to a first embodiment. The fitness equipment **200** is similar to the dumbbell (**100a**, **100b**, **100c**, **100d**, **100e** and **100f**) in structure. The difference between the fitness equipment **200** and dumbbell (**100a**, **100b**, **100c**, **100d**, **100e** and **100f**) includes: a length of the second grip rod **20** is greater than that of the first grip rod **10a**. The length of the second grip rod **20** is significantly increased, and the holding width of the user's hand can be wider, so that the fitness equipment **200** can be used as a barbell. The first grip rod (**10a**, **10b**, **10c**, **10d**, **10e** and **10f**) of the dumbbell (**100a**, **100b**, **100c**, **100d**, **100e** and **100f**), on the other hand, has a shorter length and a narrower holding width of the user's hand, which can be suitable for the usage of the dumbbell.

(81) In at least one embodiment, the second grip rod **20** has an integrally formed structure, which is a cylindrical bar.

(82) In other embodiment, the second grip rod **20** can also be a detachable assembly, for example, a snap structure, a socket structure, or a thread structure is formed on the second grip rod **20**, and the second grip rod **20** is assembled by a plurality of second grip rods **20**. During producing and selling, multiple second grip rods **20** are separated from each other to facilitate packaging and transportation; while when used by users, users can assemble multiple second grip rods **20** to form second grip rods **20** of different lengths according to the needed length of use.

(83) Referring to FIGS. **12** and **13**, the present disclosure provides a fitness equipment **200'** according to a second embodiment. The fitness equipment **200'** includes a second grip rod **20'**, and at least two dumbbells **100'** connected with both ends of the second grip rod **20'** respectively.

(84) The dumbbell **100'** is similar to the dumbbell (**100a**, **100b**, **100c**, **100e**, **100f**, **100g**, **100h**) in structure. The difference between the fitness equipment **200'** and dumbbell (**100a**, **100b**, **100c**, **100e**, **100f**, **100g**, **100h**) includes: a length of the second grip rod **20'** is greater than that of above first grip rod; two end of the second grip rod **20'** may be firmly bonded with the second counterweights **33'**.

(85) Herein, the second grip rod **20'** is longer than above first grip rod, thus allowing the user to use the fitness equipment **200'** as a barbell.

(86) In at least one embodiment, the protecting elements **50'** are sleeved on the first counterweights **31a'** farthest from the second grip rod **20'**. It should be understood that, the protecting elements **50'** can be sleeved on other first counterweights **31'** and/or the second counterweights **33'**.

(87) Referring to FIG. **14**, the present disclosure provides a fitness equipment **200''** according to a third embodiment. The fitness equipment **200''** includes a second grip rod **20''**, and at least two

dumbbells **100''** connected with both ends of the second grip rod **20''** respectively.

(88) The difference between the fitness equipment **200''** and the fitness equipment **200'** includes: the dumbbell **100''** is similar to the dumbbell **100d** in structure; the second grip rod **20''** includes two third connecting portions **21''**, the third connecting portions **21''** are received in the second screw holes **313''** of the first counterweights **31''**, to connect the second grip rod **20''** with the dumbbells **100''**.

(89) In at least one embodiment, the third connecting portion **21''** includes an external thread.

(90) In at least one embodiment, two ends of the second grip rod **20''** are detachably connected with two dumbbells **100d**.

(91) The above description is merely some embodiments. It should be noted that for one with ordinary skills in the art, improvements can be made without departing from the concept of the present disclosure, but these improvements shall fall into the protection scope of the present disclosure.

## Claims

1. A dumbbell, comprising: a first grip rod, comprising a body portion and two thread connection portions respectively disposed at two ends of the body portion; each of the two thread connection portions defines a first screw hole; wherein threads are arranged on only an inner hole wall of the first screw hole; an end surface of each of the two thread connection portions away from the body portion is recessed inwardly towards the body portion to form the first screw hole having the threads on the inner hole wall; an outer circumferential surface of each of the two thread connection portions is flushed with an outer circumferential surface of the body portion of the first grip rod; and two counterweight assemblies, each of the two counterweight assemblies comprises at least one first counterweight, the at least one first counterweight comprises a first connection portion, the first connection portion is threadedly received in the respective first screw hole and connected to the threads on the inner hole wall of the respective first screw hole to connect the first grip rod with the at least one first counterweight; wherein, each of the two counterweight assemblies further comprises a second counterweight, the second counterweight comprises a second connection portion, a second screw hole is defined in each of the at least one first counterweight, the second connection portion is threadedly received in the respective second screw hole; wherein the dumbbell further comprises at least two protecting elements, each protecting element of the at least two protecting elements is sleeved on respective first counterweight of the at least one counterweights adjacent to the first grip rod; and/or each protecting element of the at least two protecting elements is sleeved on respective first counterweight of the at least one counterweights farthest from the first grip rod.

2. The dumbbell according to claim 1, wherein no screw hole is defined in an opposite surface of each of the second counterweights away from the first grip rod.

3. The dumbbell according to claim 2, wherein each of the opposite surfaces of the second counterweights away from the first grip rod is flat.

4. The dumbbell according to claim 1, each protecting element of the at least two protecting elements is sleeved on respective first counterweight of the at least one counterweights adjacent to the first grip rod; and/or each protecting element of the at least two protecting elements is sleeved on respective second counterweight.

5. The dumbbell according to claim 4, wherein each protecting element of the at least two protecting elements comprises: a protecting portion, sleeved on a first periphery surface of the respective first counterweight or sleeved on a second periphery surface of the second counterweight; and a clamping portion, connected with the protecting portion at an angle, the clamping portion is attached on a first side surface of the respective first counterweight or attached on a third side surface of the second counterweight.

6. The dumbbell according to claim 1, further comprising a sleeve, the sleeve is sleeved on the first grip rod, and the sleeve is made of foam, rubber, or flexible plastic.
  7. The dumbbell according to claim 6, wherein the sleeve in a natural state has a thickness of 3.5~4.5 mm, and the sleeve in a compressed state has a thickness of 2~3 mm.
  8. The dumbbell according to claim 1, wherein a quantity of the at least one first counterweight is more than one, each of the more than one first counterweights defines a second screw hole, two adjacent counterweights of the more than one first counterweights are connected by screwing the first connection portion into respective second screw hole.
  9. The dumbbell according to claim 1, wherein each protecting element of the at least two protecting elements comprises: a protecting portion, sleeved on a first periphery surface of the respective first counterweight; and a clamping portion, connected with the protecting portion at an angle, the clamping portion is attached on a first side surface or a second side surface of the respective first counterweight, the first side surface is opposite to the second side surface.
  10. The dumbbell according to claim 1, wherein the dumbbell further comprises at least one first washer, and each of the at least one first washer is sleeved on the first connection portion of a respective one of the at least one first counterweight of a respective one of the two counterweight assemblies.
  11. The dumbbell according to claim 1, wherein the dumbbell further comprises at least one first washer, and each of the at least one first washer is sleeved on the second connection portion of the second counterweight of a respective one of the two counterweight assemblies.
  12. The dumbbell according to claim 1, wherein the dumbbell further comprises at least one second washer, and each of the at least one second washer is sleeved on a respective one of the at least one first counterweight of a respective one of the two counterweight assemblies.
  13. The dumbbell according to claim 1, wherein the dumbbell further comprises at least one second washer, and each of the at least one second washer is sleeved on the second counterweight of a respective one of the two counterweight assemblies.
  14. The dumbbell according to claim 1, wherein, for each of the two counterweight assemblies, each of the at least one first counterweight comprises a chamfer, the second counterweight also comprises a chamfer, after the at least one first counterweight is connected with the second counterweight, the chamfers of a respective one of the at least one first counterweight and the second counterweight cooperatively form a groove, the dumbbell further comprises a third washer, the third washer is received in the groove.
  15. The dumbbell according to claim 1, wherein a surface of the first grip rod is provided with at least one anti-slip groove.
  16. The dumbbell according to claim 1, wherein the at least one first counterweight has a weight of 0.25~1.25 Kg; and/or the at least one first counterweight has a diameter of 63~118 mm.
  17. A fitness equipment, comprising a second grip rod and at least two dumbbells as described in claim 1, wherein a length of the second grip rod is greater than a length of the first grip rod, and the at least two dumbbells are threadedly connected with two opposite ends of the second grip rod; wherein, the second grip rod is integrally formed; or, the second grip rod is assembled in a detachable manner.
  18. The dumbbell according to claim 1, wherein, each end of the first grip rod has an end surface facing towards a respective one of the two counterweight assemblies; the end surface of each end is recessed inwardly to form the first screw hole; for each of the two counterweight assemblies, each of the at least one first counterweight has a counterweight end surface facing towards the first grip rod, and the first connection portion is protruding from the counterweight end surface; for each of the two counterweight assemblies, when the first connection portion is threadly received in the respective first screw hole, a portion of the respective end surface of the first grip rod that is not recessed directly or indirectly abuts against the counterweight end surface.
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